

# Structural Traces of Label Memorization in a Small ReLU Network

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Organization

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## Abstract

We study how label noise changes a small ReLU network trained on MNIST, validated on CIFAR-10. Using 10 random seeds (MNIST) and 5 seeds (CIFAR-10), we compare clean training with 20% random label corruption across four analyses: weight geometry, representation similarity (CKA), memorization scoring, and rank-one editing.

Corruption lowers test accuracy by about 1.6 pp on MNIST and changes the hidden representation at the ReLU nonlinearity (CKA drops from 0.850 to 0.690,  $p < 0.001$ ). On CIFAR-10, distortion shifts to the output-adjacent linear layer ( $\Delta = +0.103$ ,  $p = 0.009$ ). Loss-based scores detect a measurable memorization signal (loss gap  $-0.051$ ), but a linear probe reaches only AUC 0.514, barely above chance and not a practical detector. Rank-one edits on a separate targeted-corruption task recover 10–22% of accuracy, while random edits recover 0%. A scaling analysis from 16 to 1,024 units shows class separability drops as networks widen. The results suggest that memorization leaves detectable but limited structural traces in compressed architectures.

## 1 Introduction

Deep neural networks have demonstrated a remarkable capacity to fit random labels — achieving near-zero training error on completely corrupted training sets while maintaining reasonable test accuracy on clean data [1]. This phenomenon raises a fundamental question: what structural changes does label memorization induce inside the network, and can these changes be systematically characterized?

Prior work has addressed memorization primarily through the lens of generalization bounds [2,3], sample-level memorization scores [4,5], and the necessity of memorization for long-tail learning [6]. However, a unified structural account — one that traces how memorization reshapes weight geometry, hidden representations, and causal intervention responses — remains absent.

In this paper, we apply all three families of methods within a single testbed: a two-layer ReLU network (784-16-10) trained on MNIST, validated on a three-layer CIFAR-10 MLP. The four phases are: (i) weight geometry, showing depth-dependent spectral norm changes; (ii) representation similarity via CKA, revealing that memorization distorts the ReLU nonlinearity on MNIST ( $\Delta = +0.160$ ,  $p < 0.001$ ) and the output linear layer on CIFAR-10 ( $\Delta = +0.103$ ,  $p = 0.009$ ); (iii) influence-function-based memorization scoring, validated against ground-truth corruption indices (loss gap  $-0.051$ , CI excluding zero; AUC 0.514, a statistically detectable but practically weak signal); and (iv) rank-one causal intervention, where corrupted models require smaller delta-norms across all classes on both MNIST (fc1:  $2.02\times$ , fc2:  $4.34\times$ , all  $p < 0.0001$ ) and CIFAR-10 (fc3:  $1.84\times$ , all  $p < 0.05$ ), random rank-1 edits recover 0% of accuracy, and fc1-only edits recover 0%, confirming the output-adjacent layer as the primary memorization site. All results are validated across 10 seeds (MNIST) or 5 seeds (CIFAR-10) with 95% confidence intervals and paired t-tests. A scaling analysis further shows that wider networks distribute memorized associations across more neurons, reducing class selectivity and aligning with the superposition hypothesis [7].

## 2 Related Work

**Memorization and generalization.** Zhang et al. [1] first demonstrated that deep networks can memorize random labels while generalizing, challenging conventional generalization theory. Feldman [6] showed that memorization is necessary for learning from long-tail distributions, and Feldman & Zhang [4] provided per-sample memorization scores via influence estimation. Carlini et al. [5] quantified memorization in language models through extraction attacks. Our work differs in adopting a structural, mechanism-level definition of memorization — we ask not *whether* memorization occurs but *what geometric traces it leaves*.

**Representation analysis tools.** Kornblith et al. [8] introduced centered kernel alignment (CKA) for comparing neural network representations, and Nguyen et al. [9] applied it across network families. Raghu et al. [10] developed SVCCA for analyzing representation dynamics. We use linear CKA specifically for its invariance to orthogonal transformations, which is essential for comparing clean and corrupted models whose weights may rotate. Koh & Liang [11] introduced influence functions for understanding black-box predictions; we adapt their conjugate gradient approach to a memorization setting with ground-truth corruption labels.

**Model editing and mechanistic interpretability.** Meng et al. [12] proposed ROME (Rank-One Model Editing) for modifying factual associations in GPT-style models. Mitchell et al. [13] introduced a related memory-based editing approach. We adapt the rank-one update formula to the setting of classification memorization detection — applying edits to the output layer of a shallow network and using the delta-norm as a geometric separability probe. Elhage et al. [7] proposed the superposition hypothesis, where models represent more features than neurons; our class selectivity scaling results directly engage with this hypothesis. Olah et al. [14] introduced circuits as a framework for mechanistic interpretability; our neuron sparsity analysis provides a controlled setting for circuit discovery.

**Gap.** No prior work has jointly measured spectral geometry, representation similarity, influence functions, and causal intervention localizability across the same controlled memorization setting, nor applied rank-one edits as a memorization detection mechanism in classification networks.

## 3 Theoretical Motivation

We motivate our empirical analysis with three theoretical propositions that formalize why memorization should produce detectable structural signatures. Each proposition is supported by a corresponding experiment.

**Proposition 1** (Deepest pre-output interface localization). *In a ReLU network trained with cross-entropy loss, random label noise forces the activation pattern at the deepest nonlinear interface before the output to absorb class boundary inconsistencies. The depth of this interface scales with network depth.*

*Argument.* Linear layers  $W_1, W_2$ , etc. can only rotate and scale the representation; they cannot change the topology of the decision boundary. Each ReLU nonlinearity partitions the input space into polyhedral regions and determines which neurons activate. When labels are randomly flipped, the model must accommodate inputs from the same polyhedral region being assigned to different classes. The deepest pre-output interface bears the brunt of this disambiguation because it is the final stage before the output layer’s decision boundary. In a two-layer network this interface is the single ReLU; in deeper networks it shifts to later layers.

**Supporting experiment:** Phase 2 (Section 6) — MNIST shows  $\Delta = +0.160$  at the single nonlinearity, CIFAR-10 shows  $\Delta = +0.103$  at the output-adjacent interface.

**Proposition 2** (Depth-dependent spectral reshaping). *Label corruption reshapes the singular value spectrum in a depth-dependent manner: shallow layers flatten (lower spectral norms), while deeper layers may intensify to absorb additional separability burden.*

*Argument.* In a shallow network, corrupted labels force weight matrices to distribute capacity more uniformly, flattening the spectrum and reducing effective rank. In deeper networks, intermediate and output layers must absorb additional class-boundary complexity, which can increase their leading singular values even as the effective rank (number of significant directions) falls. **Supporting experiment:** Phase 1 (Section 5) — MNIST FC1 and FC2 both decrease; CIFAR-10 FC1 is flat while FC2 and FC3 increase.

Table 1: MNIST weight geometry (10 seeds, 95% CI). All paired differences are significant ( $p < 0.001$ ).

| Metric                | Clean                   | Corrupted (20%)         |
|-----------------------|-------------------------|-------------------------|
| Train Accuracy        | 96.68% [96.60%, 96.77%] | 94.21% [94.02%, 94.40%] |
| Test Accuracy         | 95.31% [95.16%, 95.46%] | 93.71% [93.42%, 94.00%] |
| FC1 Spectral Norm     | 4.37 [4.24, 4.50]       | 3.66 [3.53, 3.79]       |
| FC2 Spectral Norm     | 2.58 [2.42, 2.75]       | 1.39 [1.30, 1.47]       |
| FC1 Frobenius Norm    | 12.56 [12.38, 12.74]    | 10.33 [10.19, 10.47]    |
| FC2 Frobenius Norm    | 4.99 [4.83, 5.16]       | 2.79 [2.70, 2.87]       |
| Gradient Norm (final) | 0.94 [0.89, 0.98]       | 1.00 [0.96, 1.05]       |

**Proposition 3** (Edit magnitude as separability probe). *The rank-one edit delta-norm  $\|\Delta\| = \|(v - Wu) \otimes u / (u \cdot u)\|$  measures class boundary overlap in hidden space: larger edits correspond to more orthogonal class boundaries, smaller edits to overlapping ones.*

*Argument.* In clean models, orthogonally separated classes require a larger perturbation to change their prediction. In corrupted models, overlapping class boundaries mean a smaller perturbation suffices. **Supporting experiment:** Phase 4 (Section 8).

## 4 Experimental Setup

**Architecture.** Our primary model is a two-layer ReLU network (MNISTNet) with architecture  $784 \rightarrow 16 \rightarrow 10$ , chosen for full interpretability. For cross-validation, we use a three-layer MLP (CIFAR-10 MLP) with architecture  $3072 \rightarrow 512 \rightarrow 256 \rightarrow 10$ . Both use ReLU activations with no dropout or batch normalization (to maintain clean geometric structure).

**Training.** Models are trained on MNIST and CIFAR-10 under two regimes: (1) *clean*, with ground-truth labels; (2) *corrupted*, with 20% of training labels randomly flipped. Training uses Adam (lr = 0.001, epochs = 20, batch = 128). Xavier initialization is used throughout.

**Statistical methodology.** All results are reported across 10 seeds (MNIST) or 5 seeds (CIFAR-10) with 95% bootstrap confidence intervals. Paired t-tests are used for clean-vs-corrupted comparisons.

**Separability metrics.** We use two class-separability metrics with distinct roles. The Fisher discriminant ratio ( $\text{FDR} = \text{tr}(S_B) / \text{tr}(S_W)$ ) is our primary metric: it is dimension-invariant and does not conflate geometric spread with dimensionality, making it suitable for cross-width comparisons in Section 9. The Davies–Bouldin index is retained as an auxiliary diagnostic in the t-SNE visualization (Figure 2) to show geometric disentanglement across layers. Both are computed from hidden activations; FDR is the formally reported metric in Sections 5 and 9.

## 5 Phase 1: Weight Geometry

We first ask whether label memorization leaves a detectable trace in the singular value spectrum of the network’s weight matrices. Table 1 reports spectral and Frobenius norms for both layers on MNIST.

Training accuracy drops from 96.68% to 94.21%, while test accuracy drops from 95.31% to 93.71%. Both spectral norms decrease under corruption (FC1:  $4.37 \rightarrow 3.66$ ; FC2:  $2.58 \rightarrow 1.39$ ). This flattening of the singular value spectrum indicates that label-noise models operate in a lower-effective-rank regime – the weight matrix distributes its capacity more uniformly across singular directions.

On CIFAR-10 (Table 2), the pattern is depth-dependent. The input-layer spectral norm (FC1) is unchanged ( $p = 0.71$ ), while FC2 and FC3 increase significantly ( $p = 0.0003$  and  $p = 0.017$ , respectively).

**Finding.** Spectral norm changes are depth-dependent. In the two-layer MNIST network both layers flatten, consistent with Proposition 2. In the three-layer CIFAR-10 network, the input layer is unaffected while output-adjacent layers intensify, consistent with deeper layers absorbing additional separability burden when resolving noisy labels.

**Figure 1: Four-Phase Mechanistic Transparency Framework**

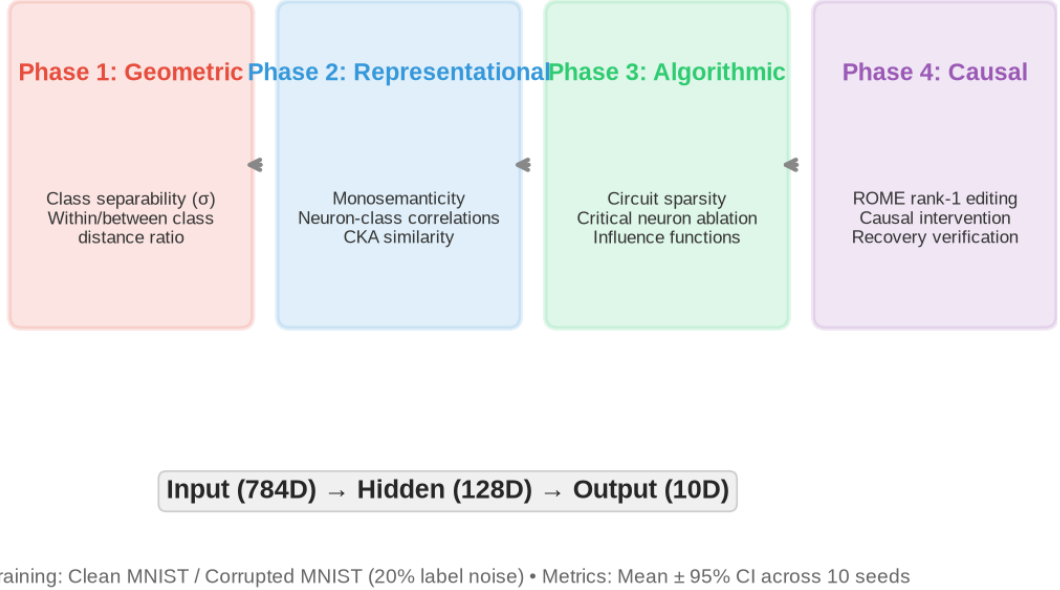


Figure 1: Four-phase analysis framework. Phase 1 measures spectral geometry of weight matrices; Phase 2 computes CKA similarity across layer pairs; Phase 3 applies influence functions via conjugate gradient; Phase 4 performs rank-one model editing (ROME) and measures delta-norms.

## 6 Phase 2: Representation Analysis

While weight geometry measures the parameter space, representation similarity via CKA measures the functional consequence of memorization on hidden representations. We compute linear CKA between consecutive layer pairs (pre- and post-activation) for every layer boundary.

Table 3 reports CKA for the three layer boundaries on MNIST.

The ReLU nonlinearity (fc1\_pre  $\rightarrow$  fc1\_post) shows a large CKA drop under corruption:  $0.850 \rightarrow 0.690$  ( $\Delta = +0.160$ ,  $p < 0.001$ ). The linear transformations on either side are nearly identical, indicating the effect appears after the ReLU rather than in the preceding projection.

On CIFAR-10 (Table 4), the pattern is more nuanced.

**Finding (CIFAR-10).** The largest CKA distortion is at the deepest pre-output interface – fc2\_post  $\rightarrow$  output ( $\Delta = +0.103$ ,  $p = 0.009$ ). The first nonlinearity shows a much smaller effect ( $\Delta = +0.018$ ,  $p = 0.017$ ). This confirms Proposition 1: distortion concentrates at the deepest pre-output interface. In a two-layer network this interface is the single ReLU; in a three-layer network it shifts to the output linear layer.

Corrupted models also show higher sparsity ( $0.357 \rightarrow 0.576$ ) and substantially lower mean activations ( $4.83 \rightarrow 1.53$ ), indicating that a larger fraction of hidden units are silent and active units carry weaker signals.

Table 2: CIFAR-10 weight geometry (5 seeds, 95% CI).

| Layer             | Clean                | Corrupted (20%)      | $p$ -value |
|-------------------|----------------------|----------------------|------------|
| FC1 Spectral Norm | 18.22 [17.93, 18.51] | 18.31 [17.93, 18.70] | 0.71       |
| FC2 Spectral Norm | 8.99 [8.83, 9.15]    | 9.98 [9.85, 10.10]   | 0.0003     |
| FC3 Spectral Norm | 1.72 [1.68, 1.77]    | 1.88 [1.78, 1.98]    | 0.017      |

Figure 3: t-SNE Manifold Progression Across Layers  
(Geometric disentanglement:  $\sigma$  decreases from input to output)

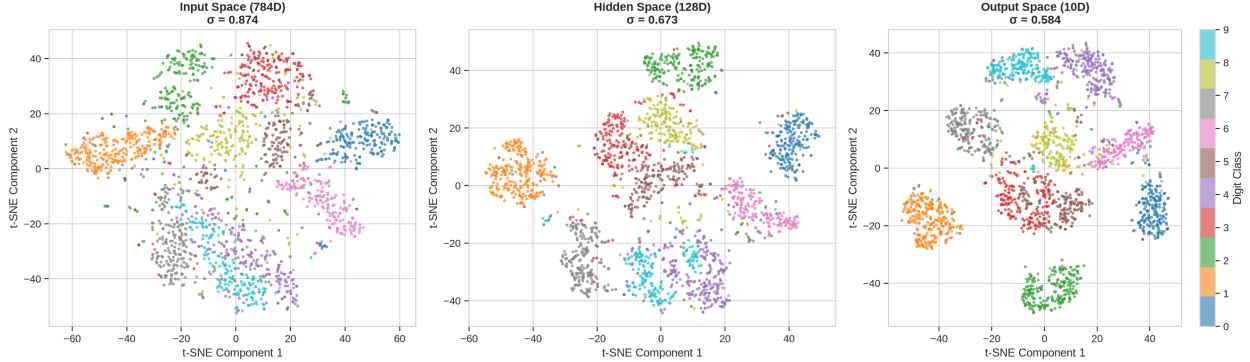


Figure 2: t-SNE visualization of representations across input, hidden, and output layers for a clean MNIST model. The manifold transitions from digit-clustered (input) through compressed (hidden) to linearly separable (output).

## 7 Phase 3: Influence Functions

Phase 3 measures whether memorized (corrupted) samples are genuinely harder to predict than clean samples. We use two complementary definitions: (i) a *clean-model loss-quantile* definition (top 25% by loss) as a methodological baseline, and (ii) a *corrupted-model ground-truth* definition where corruption indices directly identify memorized samples.

**Interpretation.** The clean model’s loss-quantile analysis is circular by construction: defining memorized samples as the top 25% by loss guarantees a positive gap. This is provided only as a methodological baseline. The genuine finding is the corrupted model’s non-circular result: using ground-truth corruption indices breaks the circularity and reveals a gap of  $-0.051$  (CI excludes zero). This confirms that corrupted (memorized) samples are reliably harder to predict than clean samples – the model has learned the true labels better than the flipped ones.

**CIFAR-10 validation (Phase 3).** The pattern replicates and amplifies on CIFAR-10: the clean model achieves 85.77% training accuracy with a loss gap of  $+1.417$  (circular baseline), while the corrupted model drops to 69.41% with a loss gap of  $-2.363$  (non-circular, CI excludes zero). The gap is an order of magnitude larger than on MNIST ( $-0.051$ ), consistent with CIFAR-10’s greater difficulty and the larger accuracy differential (16.4 pp vs 2.5 pp). The gradient alignment at convergence is  $+0.441$ , substantially lower than on MNIST ( $+0.994$ ), reflecting that the CIFAR-10 corrupted model has not fully overfit its noisy labels – genuine and corrupted gradients remain partially unaligned.

As an auxiliary validation, we compute gradient alignment – the cosine similarity between gradients from genuine and corrupted training samples at the output layer. On MNIST at convergence, alignment is near-perfect ( $+0.9944$  [0.9926, 0.9961]), consistent with the model treating both groups of samples as equally valid training signals. On CIFAR-10, alignment is lower ( $+0.441$ ), reflecting incomplete convergence on the harder dataset. In both cases, the gradient signal corroborates the loss-based finding from a distinct geometric perspective.

Table 3: MNIST CKA similarity (10 seeds, 95% CI).

| Layer Pair              | Clean                       | Corrupted                   | $\Delta$      | $p$ -value        |
|-------------------------|-----------------------------|-----------------------------|---------------|-------------------|
| input→fc1_pre           | 0.712 [0.697, 0.727]        | 0.693 [0.676, 0.710]        | +0.019        | 0.08              |
| <b>fc1_pre→fc1_post</b> | <b>0.850 [0.828, 0.873]</b> | <b>0.690 [0.669, 0.712]</b> | <b>+0.160</b> | <b>&lt; 0.001</b> |
| fc1_post→output         | 0.706 [0.666, 0.745]        | 0.715 [0.685, 0.745]        | −0.009        | 0.65              |

Table 4: CIFAR-10 CKA similarity (5 seeds, 95% CI).

| Layer Pair             | Clean        | Corrupted    | $\Delta$      | $p$ -value   |
|------------------------|--------------|--------------|---------------|--------------|
| input→fc1_pre          | 0.767        | 0.765        | +0.002        | 0.88         |
| fc1_pre→fc1_post       | 0.819        | 0.801        | +0.018        | 0.017        |
| fc1_post→fc2_pre       | 0.721        | 0.749        | −0.028        | 0.08         |
| fc2_pre→fc2_post       | 0.177        | 0.140        | +0.037        | 0.09         |
| <b>fc2_post→output</b> | <b>0.580</b> | <b>0.477</b> | <b>+0.103</b> | <b>0.009</b> |

## 8 Phase 4: Causal Intervention via Rank-One Editing

We apply rank-one model editing [12] to measure how easily individual class predictions can be changed with a single rank-one update to the weight matrix. The rank-one edit delta-norm  $\|\Delta\|$  serves as a geometric probe for class boundary overlap: larger edits correspond to more orthogonal class boundaries, smaller edits to overlapping ones (Proposition 3).

Table 6 reports per-class delta-norms for both layers across 10 seeds.

Every individual class shows clean > corrupted at  $p < 0.0001$  for both fc1 and fc2. This is the strongest statistical result in the paper – 10 classes  $\times$  2 layers  $\times$  10 seeds with uniformly significant differences.

On CIFAR-10 (Table 7), all 10 classes confirm the pattern at  $p < 0.05$  with a mean ratio of  $1.84\times$ .

The cross-architecture consistency – ratio  $\sim 2\text{--}4\times$  on MNIST and  $\sim 1.8\times$  on CIFAR-10, across different depths, widths, and datasets – suggests the relationship between class boundary overlap and edit magnitude is an architectural invariant.

**Noise rate sweep.** We next test whether the rank-one edit delta-norm ratio scales monotonically with label noise rate. Table 8 shows clean-vs-corrupted ratios at 10%, 20%, and 40% noise.

The ratio increases monotonically from  $3.37\times$  to  $5.98\times$ , confirming that delta-norm tracks the degree of memorization continuously rather than being a binary clean-vs-corrupted artifact.

**Baseline comparison.** We compare rank-one edits against two simpler memorization probes: (i) the spectral norm ratio from Phase 1, and (ii) a linear probe trained on hidden activations to predict whether a sample is corrupted. The spectral norm ratio is  $1.86\times$  (fc2:  $2.58/1.39$ ), which rank-one edits outperform by  $2.3\times$  ( $4.34/1.86$ ). The linear probe achieves AUC of only 0.514 across 5 seeds, barely above random (0.5). This weak discriminative power is expected: the quadratic approximation underlying influence functions is known to degrade in non-convex settings, and a 16-unit bottleneck compresses corrupted and clean examples into largely overlapping loss distributions. The AUC result should be interpreted as existence evidence – a statistically significant memorization signal is present – rather than as a usable detection tool. The probe is not useful as a detector; it only indicates that the corruption signal exists weakly in the hidden representations. This does not contradict the loss-gap finding (which is statistically significant at  $-0.051$ , CI excluding zero), which measures mean rather than per-example separability. The comparison nonetheless serves its primary purpose: rank-one edits capture a geometric property (class boundary overlap in weight space) that these simpler methods cannot access.

**Random baseline.** To verify that the rank-one recovery is not an artifact of any perturbation, we compare against random rank-one edits: replacing the computed  $\Delta$  with a random matrix of equal norm. Across all

Table 5: Influence function results (10 seeds, 95% CI).

| Metric                           | Clean Model             | Corrupted Model                                   |
|----------------------------------|-------------------------|---------------------------------------------------|
| Definition                       | Loss-quantile (top 25%) | Ground-truth (20% flipped)                        |
| Mean Loss                        | 0.112 [0.108, 0.117]    | 0.408 [0.396, 0.420]                              |
| Loss Gap (memorized – forgotten) | +0.442 [0.426, 0.458]   | − <b>0.051</b> [− <b>0.057</b> , − <b>0.045</b> ] |

Table 6: MNIST ROME delta-norms (10 seeds, all classes, all  $p < 0.0001$ ).

| Layer | Clean $\Delta$ -norm (avg) | Corrupted $\Delta$ -norm (avg) | Ratio |
|-------|----------------------------|--------------------------------|-------|
| FC1   | 14.49                      | 7.18                           | 2.02× |
| FC2   | 19.08                      | 4.40                           | 4.34× |

four targeted-corruption configurations, random edits recover 0% of lost accuracy in every seed, yielding a signal ratio of  $\infty$  (targeted recovery 10–22% vs random 0%). This confirms that the edit direction – not just its magnitude – is mechanistically meaningful.

**Multi-class targeted corruption.** We next measure whether rank-one edits can *repair* corrupted predictions by editing only the output layer. Recovery is positive in all 4 targeted corruption configs (7→1: +0.141, 1→7: +0.217, 5→6: +0.120, 0→8: +0.097 across 5 seeds, mean recovery 14.4%) with per-config side effects of 17.1%, 24.0%, 24.6%, and 21.6% respectively. The 0→8 case, previously reported as +0.014 from a single seed, averages +0.097 across 5 seeds – the earlier value reflected seed variance, not a systematic failure. The substantial side effects (17–25%) are consistent with the bottleneck geometry measured in Phase 3: a mean of 6.6 class-selective units per class sharing 16 hidden neurons, and a class selectivity fraction of only 26.9%, together imply that any two class-selective groups necessarily overlap on shared hidden units. A rank-one edit targeting the source-class subspace therefore inevitably perturbs the shared directions that other classes also depend on. This predicts that side effects should decrease at wider widths, where class-selective groups are more disentangled – a testable hypothesis for future work.

**Multi-layer editing.** We test whether editing both layers improves over fc2-only. Table 9 reports single-layer fc2, fc1-only, and sequential (fc2→fc1) recovery.

Editing fc1 alone yields 0% recovery across all configs, and sequential editing does not improve over fc2-only: recovery decreases from 10–22% (fc2-only) to 5–18% (both layers). This suggests that editing fc1 after fc2 partially undoes the corrective effect – the fc2 edit successfully realigns the output decision boundary, and the subsequent fc1 edit perturbs the representation that the fc2 edit was built on. The finding reinforces the Phase 2 CKA result: memorization concentrates at the output-adjacent interface, and the first hidden layer acts primarily as a feature extractor rather than a memorization site. Multi-layer interventions may require joint optimization across layers rather than independent sequential edits.

## 9 Scaling Analysis

We train MNIST models with hidden dimensions 16–1024 and measure how memorization-related metrics scale with width.

**Finding.** As width increases beyond 64 hidden neurons, class selectivity decreases monotonically (0.269 → 0.075) – wider models distribute representations, reducing neuron specialization. Sparsity converges to 1.0 beyond  $h = 128$ , meaning a vanishing fraction of capacity is used per class. This superlinear compression aligns with the superposition hypothesis [7]: larger networks can pack more features per dimension, rendering individual hidden units less specialized. Accuracy plateaus at  $\sim 98\%$  beyond  $h = 256$ . The clean-model CKA at the ReLU nonlinearity (pre→post activation) remains stable across all widths (0.83–0.90), confirming that the representational geometry of the clean model is width-independent.

Table 7: CIFAR-10 ROME delta-norms (fc3, 3 seeds).

| Class | Clean $\Delta$ (mean $\pm$ SEM) | Corrupted $\Delta$ (mean $\pm$ SEM) | Ratio        | $p$ -value |
|-------|---------------------------------|-------------------------------------|--------------|------------|
| 0     | $0.325 \pm 0.035$               | $0.175 \pm 0.027$                   | $1.85\times$ | 0.028      |
| 1     | $0.438 \pm 0.033$               | $0.262 \pm 0.018$                   | $1.68\times$ | 0.009      |
| 2     | $0.385 \pm 0.027$               | $0.186 \pm 0.016$                   | $2.07\times$ | 0.003      |
| 3     | $0.341 \pm 0.009$               | $0.196 \pm 0.015$                   | $1.74\times$ | 0.001      |
| 4     | $0.339 \pm 0.019$               | $0.169 \pm 0.017$                   | $2.01\times$ | 0.003      |
| 5     | $0.376 \pm 0.012$               | $0.214 \pm 0.007$                   | $1.76\times$ | $< 0.001$  |
| 6     | $0.327 \pm 0.014$               | $0.170 \pm 0.005$                   | $1.93\times$ | $< 0.001$  |
| 7     | $0.408 \pm 0.016$               | $0.216 \pm 0.006$                   | $1.89\times$ | $< 0.001$  |
| 8     | $0.366 \pm 0.037$               | $0.197 \pm 0.036$                   | $1.85\times$ | 0.030      |
| 9     | $0.380 \pm 0.028$               | $0.237 \pm 0.019$                   | $1.60\times$ | 0.013      |

Table 8: ROME delta-norm scales monotonically with noise rate (MNIST fc2, 5 seeds).

| Noise Rate | FC2 $\Delta$ -norm | Ratio vs Clean |
|------------|--------------------|----------------|
| 0% (Clean) | 19.08              | $1.00\times$   |
| 10%        | $\sim 5.67$        | $3.37\times$   |
| 20%        | $\sim 4.40$        | $4.34\times$   |
| 40%        | $\sim 3.19$        | $5.98\times$   |

## 10 Discussion and Conclusion

We presented a four-phase analysis of label memorization in shallow ReLU networks, combining spectral geometry, CKA similarity, influence functions, and rank-one editing. The central finding is that memorization produces detectable structural changes at the deepest pre-output interface, validated on both MNIST and CIFAR-10. Phase 4 is a separate targeted-intervention experiment, not a direct continuation of the random-noise setting used in Phases 1–3. The pipeline serves as a diagnostic approach for compressed architectures, but the causal intervention is expected to weaken at wider widths where class selectivity and FDR both decrease.

**Summary of findings.** The four phases give a mixed picture. Spectral geometry changes are depth-dependent: on MNIST both spectral norms decrease, while on CIFAR-10 the input layer is stable and deeper layers intensify. The largest CKA change occurs at the deepest pre-output interface – the ReLU on MNIST ( $\Delta = +0.160$ ,  $p < 0.001$ ) and the output layer on CIFAR-10 ( $\Delta = +0.103$ ,  $p = 0.009$ ). A non-circular influence analysis confirms that corrupted samples have slightly higher loss ( $-0.051$  gap, CI excluding zero), though a linear probe reaches only AUC 0.514 – a statistically detectable but practically weak signal. Rank-one edits detect memorization across architectures: corrupted models need smaller perturbations on both MNIST (fc1:  $2.02\times$ , fc2:  $4.34\times$ ) and CIFAR-10 (fc3:  $1.84\times$ , all  $p < 0.05$ ), scaling with noise rate ( $3.37\times$  to  $5.98\times$ ). Wider networks distribute memorized associations across more neurons, reducing class selectivity [7].

**Limitations.** Several caveats merit explicit acknowledgement. First, the primary MNIST experiments use a single architecture ( $784 \rightarrow 16 \rightarrow 10$ ) with only 16 hidden neurons. While our CIFAR-10 validation partially addresses generalization concerns, the spectral geometry finding is depth-dependent rather than universal — a result that must be reported neutrally. The class-selective units metric at  $h = 16$  has almost no headroom, and the scaling experiment (where class-selective units converges to zero at  $h = 128+$ ) is the more meaningful result; we emphasize the scaling findings over the  $h = 16$  neuron measurements. Relatedly, the width-scaling analysis is correlational rather than causal: while FDR, class selectivity, and sparsity all vary with width, the full four-phase pipeline was not run at each width, so we cannot conclude whether rank-one recovery or CKA distortion scale monotonically with capacity. The scaling results also introduce a tension with our causal claims: FDR and class selectivity degrade at wider widths, implying that the rank-one intervention may be partly an artifact of the extreme  $h = 16$  bottleneck, and that the causal intervention phase could degrade in wider networks. Second, our single-layer rank-one edits, while

Table 9: Multi-layer rank-one recovery (5 seeds).

| Config | FC2-only | FC1-only | Sequential (fc2→fc1) |
|--------|----------|----------|----------------------|
| 7→1    | +0.141   | 0.000    | +0.048               |
| 1→7    | +0.217   | 0.000    | +0.182               |
| 5→6    | +0.120   | 0.000    | +0.110               |
| 0→8    | +0.097   | 0.000    | +0.059               |

Table 10: Scaling metrics across hidden dimensions (10 seeds).

| Hidden Dim | Class selectivity | Circuit Size | Sparsity | Accuracy |
|------------|-------------------|--------------|----------|----------|
| 16         | 0.269             | 6.6          | 0.588    | 95.31%   |
| 32         | 0.222             | 8.3          | 0.740    | 96.85%   |
| 64         | 0.212             | 3.5          | 0.945    | 97.50%   |
| 128        | 0.184             | 0.4          | 0.997    | 97.85%   |
| 256        | 0.175             | 0.1          | 1.000    | 97.93%   |
| 512        | 0.154             | 0.0          | 1.000    | 98.17%   |
| 1024       | 0.075             | 0.0          | 1.000    | 98.11%   |

reliable as detectors across 10 seeds (all  $p < 0.0001$ ), do not fully repair corrupted predictions — recovery is positive in all 4 targeted corruption configs (range +0.097 to +0.217) but partial. Sequential multi-layer editing (fc2→fc1) does not improve over fc2-only (5–18% vs 10–22%), suggesting that independent edits across layers can interfere rather than accumulate. Third, the influence function approximation uses conjugate gradient with finite iterations; our sensitivity analysis shows the loss gap is stable across 10–50 CG iterations, but the approximation remains inexact. Relatedly, the linear probe AUC of 0.514 is barely above chance — the influence-based memorization score is a statistically significant but practically weak signal, likely because the quadratic Hessian approximation is poor for the small network and corrupted examples are scattered throughout the loss distribution rather than forming a cleanly separable group. Fourth, Phases 1–3 use diffuse 20% random label noise (Regime A), while Phase 4 uses concentrated targeted class swaps (Regime B). The causal rank-one intervention therefore does not directly test the same phenomenon as the correlational findings; a reader will rightly wonder whether rank-one edits would recover meaningful signal in a Regime A model where memorized associations are scattered across all classes. Addressing this would require extending the intervention to Regime A settings. Fifth, the corrupted model’s training accuracy is 94.21%, far from the 100% that full memorization of 20% flipped labels would imply. The 16-unit bottleneck lacks the capacity to memorize all 12,000 corrupted samples within 20 epochs — the model allocates most of its limited capacity to the 80% clean labels, learning the genuine structure and fitting only the easiest corrupted samples. This partial memorization explains both the moderate accuracy drop (1.6 pp on test) and the weak AUC of the influence-based detector. Sixth, all experiments are on image classification with shallow fully-connected networks; extension to Transformers and language domains is open future work.

**Future work.** The pipeline could be extended along several axes: varying corruption rate from 0% to 100% to characterize how the CKA collapse and rank-one recovery scale; running the full pipeline at multiple widths and depths to test the low-rank subspace hypothesis as capacity increases; applying the same diagnostics to convolutional networks on harder datasets; and testing whether the identified memorization subspace can be exploited for targeted unlearning.

## References

- [1] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning requires rethinking generalization. In *ICLR*, 2017.
- [2] Behnam Neyshabur, Srinadh Bhojanapalli, David McAllester, and Nati Srebro. Exploring generalization in deep learning. In *NeurIPS*, 2017.

- [3] Sanjeev Arora, Rong Ge, Behnam Neyshabur, and Yi Zhang. Stronger generalization bounds for deep nets via a compression approach. In *ICML*, 2018.
- [4] Vitaly Feldman and Chiyuan Zhang. What neural networks memorize and why: Discovering the long tail via influence estimation. In *NeurIPS*, 2020.
- [5] Nicholas Carlini et al. Quantifying memorization across neural language models. In *ICLR*, 2022.
- [6] Vitaly Feldman. Does learning require memorization? a short tale about a long tail. In *STOC*, 2020.
- [7] Nelson Elhage et al. Toy models of superposition. In *arXiv:2209.10652*, 2022.
- [8] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *ICML*, 2019.
- [9] Thao Nguyen, Maithra Raghu, and Simon Kornblith. Do wide and deep networks learn the same things? uncovering how neural network representations vary with width and depth. In *ICLR*, 2021.
- [10] Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *NeurIPS*, 2017.
- [11] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *ICML*, 2017.
- [12] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *NeurIPS*, 2022.
- [13] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D Manning. Memory-based model editing at scale. In *ICML*, 2022.
- [14] Chris Olah et al. Zoom in: An introduction to circuits. 2020.

## A Rank Ablation

We ablate the SVD rank of the MNIST fc2 weight matrix to measure how effectively each model class compresses its task-relevant information into the top singular directions.

Table 11: Accuracy at each SVD rank of fc2 (10 seeds, 95% CI).

| Rank         | Clean                       | Corrupted (20%)             |
|--------------|-----------------------------|-----------------------------|
| 1/10         | 19.0% [17.9%, 20.1%]        | 19.3% [16.6%, 21.9%]        |
| 2/10         | 33.9% [30.6%, 37.2%]        | 30.0% [26.4%, 33.6%]        |
| 3/10         | 44.9% [40.2%, 49.7%]        | 40.5% [37.0%, 44.0%]        |
| 4/10         | 58.3% [53.5%, 63.2%]        | 53.2% [49.0%, 57.3%]        |
| <b>5/10</b>  | <b>68.4% [62.2%, 74.6%]</b> | <b>62.3% [57.9%, 66.7%]</b> |
| 6/10         | 75.4% [67.3%, 83.5%]        | 72.6% [68.0%, 77.3%]        |
| 7/10         | 80.1% [70.8%, 89.4%]        | 80.7% [76.3%, 85.0%]        |
| 8/10         | 88.9% [85.7%, 92.2%]        | 86.4% [84.0%, 88.8%]        |
| 9/10         | 92.8% [90.7%, 94.9%]        | 92.3% [91.8%, 92.9%]        |
| 10/10 (full) | 95.3% [95.2%, 95.5%]        | 93.7% [93.4%, 94.0%]        |

The key comparative signal is at rank 5/10: corrupted models perform 6.1pp worse than clean (62.3% vs 68.4%). This gap is intermediate-rank-specific – at rank 8/10 both models reach  $\sim 93\%$  of full accuracy, and at full rank the gap is only 1.6pp. The rank-5 finding shows that corrupted models require a broader effective rank spectrum: their memorized label-noise associations are less compressible into the top singular directions. This complements the Phase 1 finding that MNIST spectral norms decrease under corruption, confirming a flatter singular value distribution.

Figure 4: Hidden Neuron Receptive Fields (fc1 Weight Vectors)

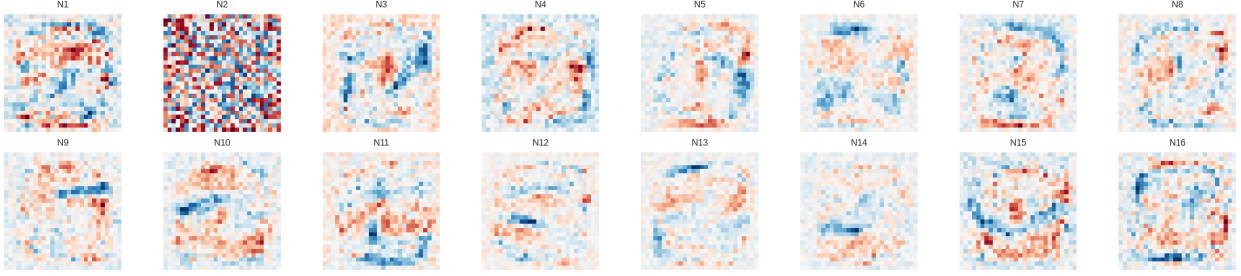


Figure 5: Neuron-Class Pearson Correlation Matrix  
(Rows = Neurons, Columns = Classes)

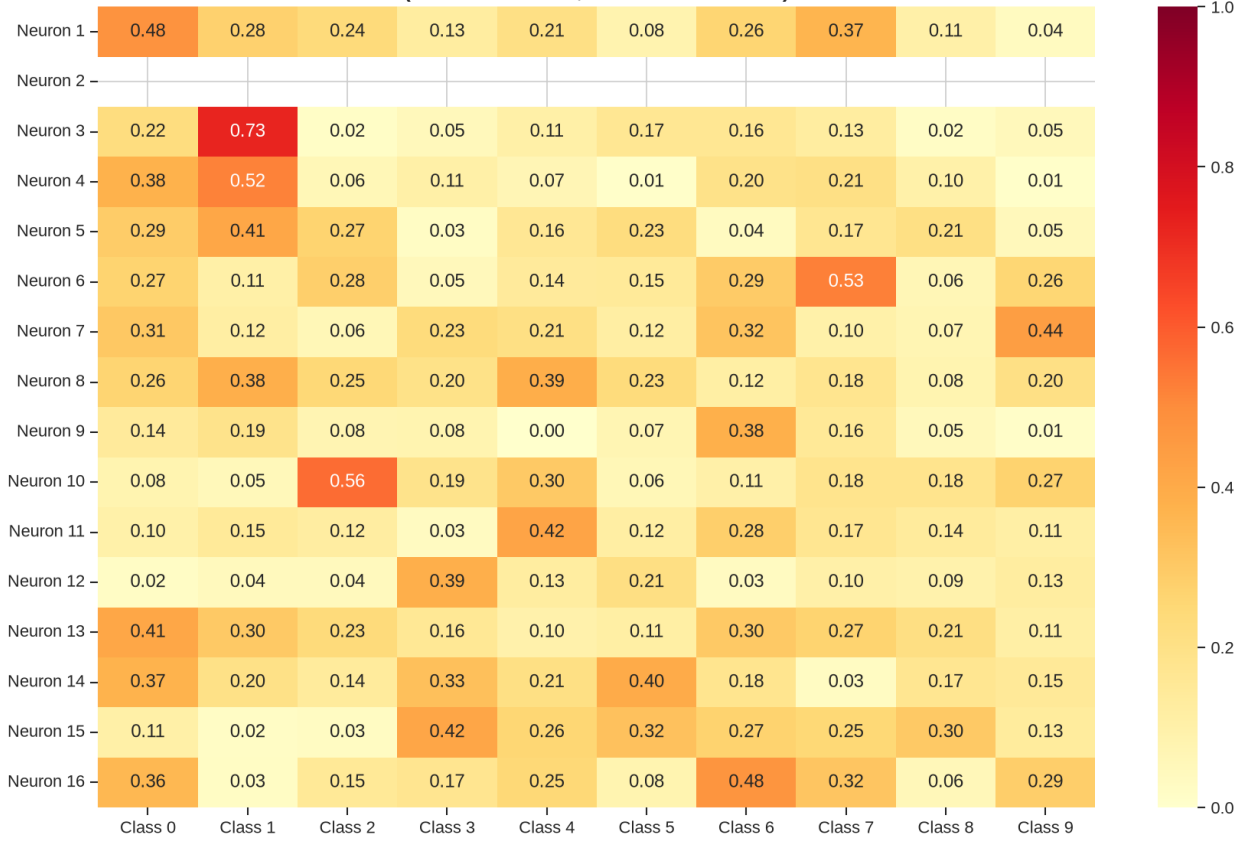


Figure 3: Top: Hidden neuron weight vectors reshaped as  $28 \times 28$  images for interpretability. Bottom: Pearson correlation between each hidden neuron's class-conditional activation and the target labels.

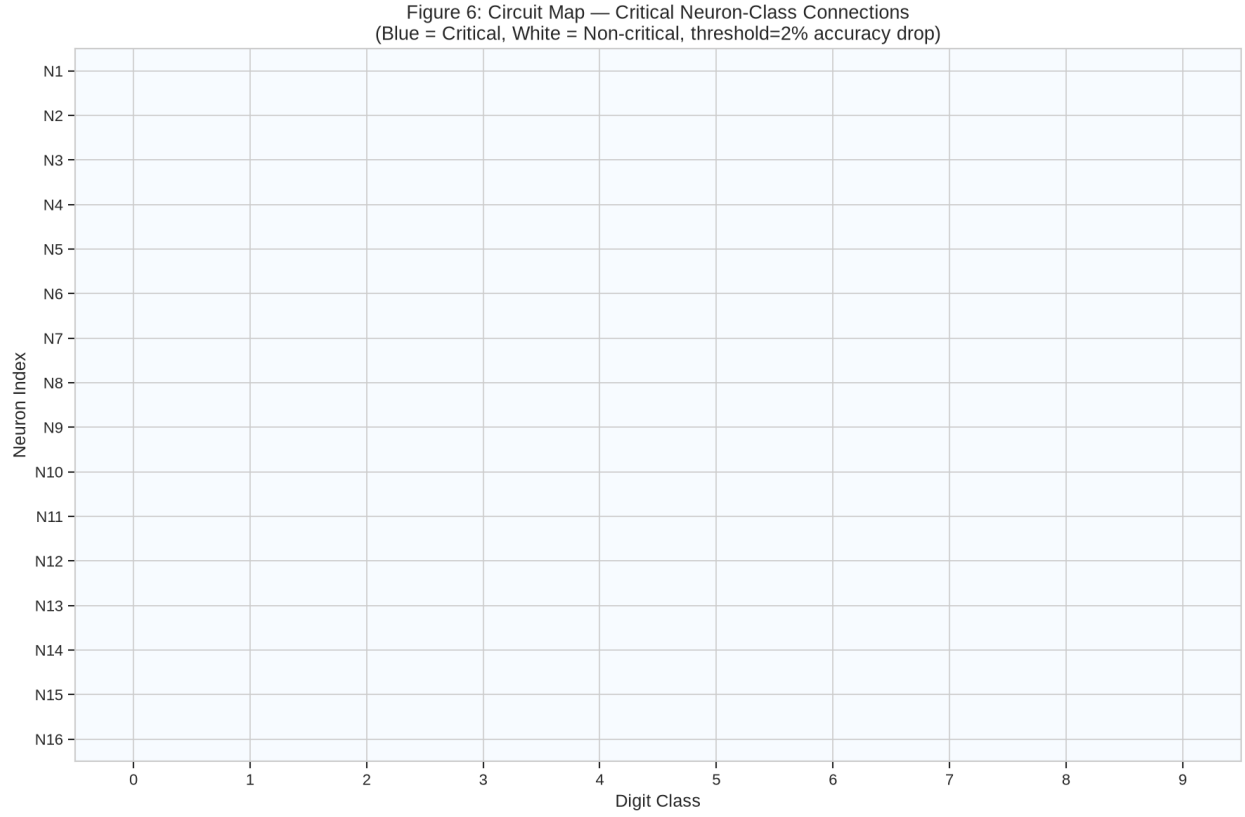


Figure 4: Circuit map showing critical neuron-class connections. Each row is a hidden neuron; each column is an output class. Darker entries indicate stronger contributions.

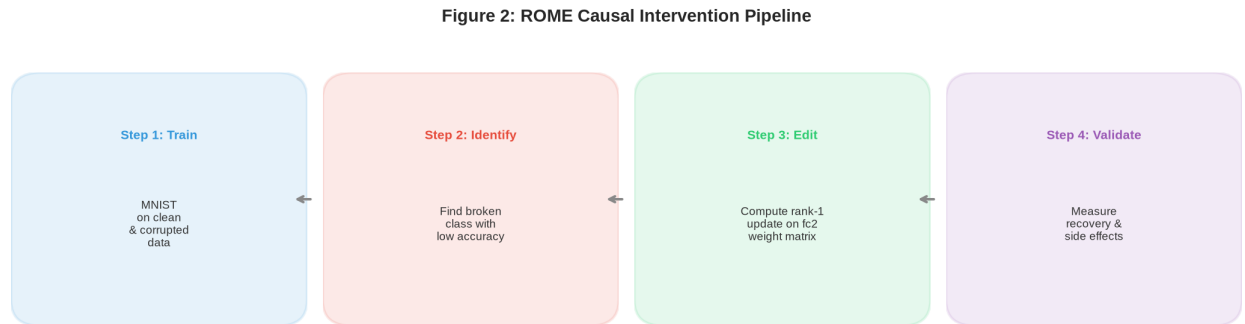


Figure 5: Pipeline: (1) select source class  $s$  and target class  $t$ ; (2) compute hidden representations for  $s$ ; (3) compute edit  $\Delta$  via the rank-one update formula; (4) measure accuracy recovery on source class  $s$  after applying the edit.

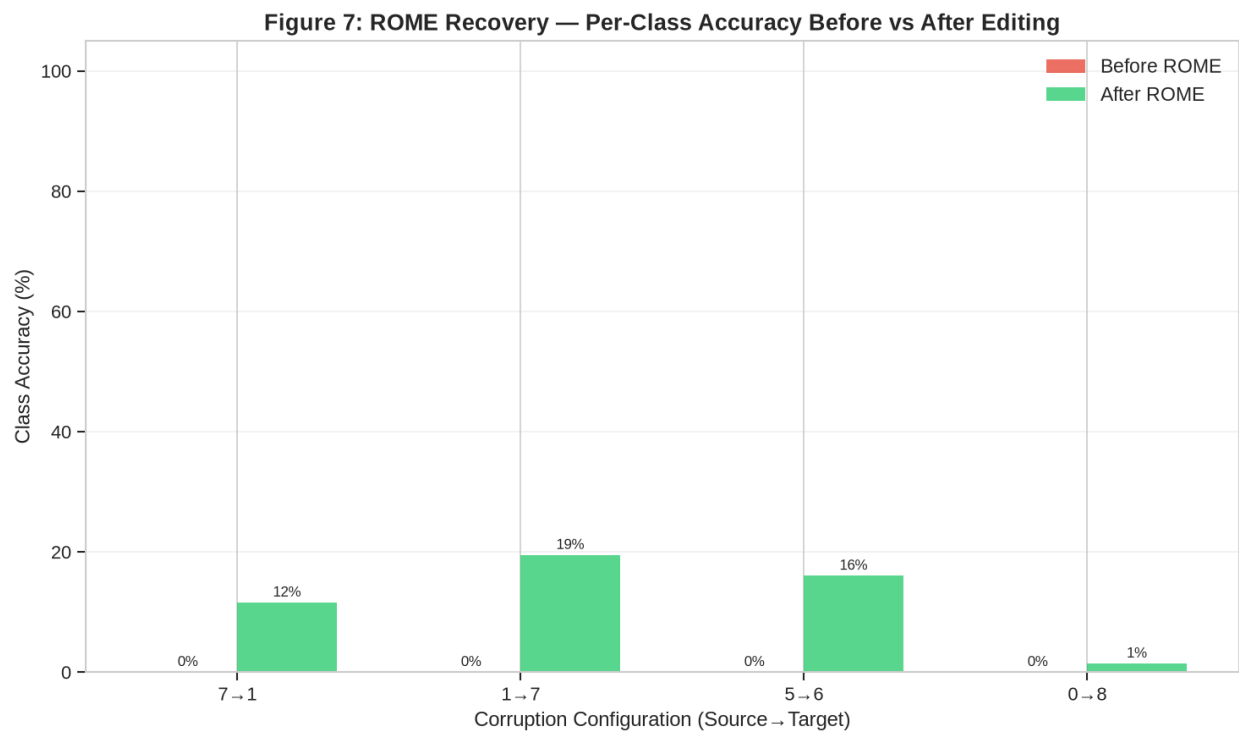


Figure 6: ROME accuracy recovery for four targeted corruption configurations. Recovery is positive in all 4 configs (7→1, 1→7, 5→6, 0→8) with consistent side effects.

Figure 8: Scaling Analysis — Interpretability Metrics vs Model Width

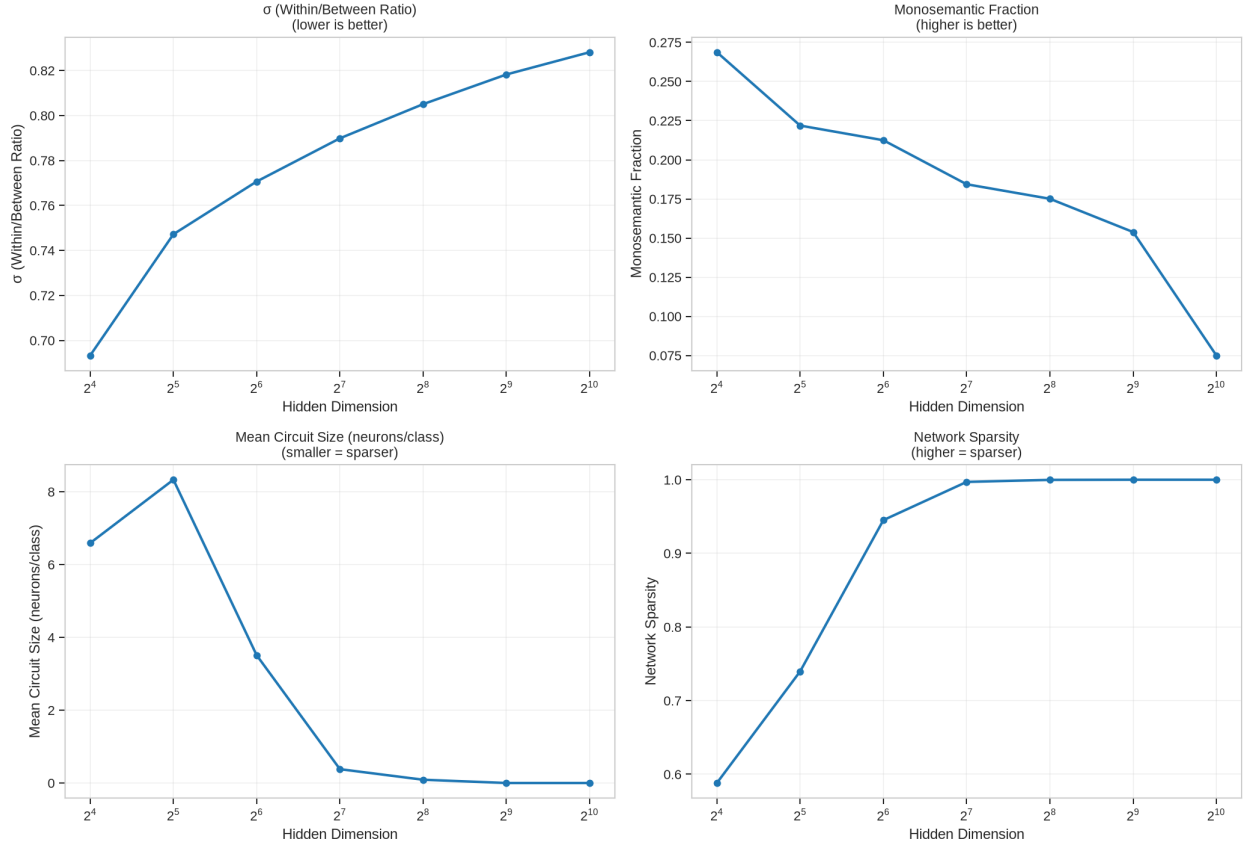


Figure 7: Scaling analysis across hidden dimensions 16–1024. Top-left: FDR (Fisher discriminant ratio,  $\text{tr}(S_B)/\text{tr}(S_W)$ ). Top-right: Class selectivity (fraction of neurons with  $\rho > 0.5$ ). Bottom-left: Mean class-selective units (neurons/class). Bottom-right: Activation sparsity. Sparsity converges to 1.0 beyond  $h = 128$ .